**User:**

**Background:** Platoon coordination is a type of problem that studies how to coordinate vehicles to form a platoon on a traffic network topology map. In this issue, vehicles need to cooperate with each other, exchange information with each other to form a platoon as much as possible at hubs or on the road.
**Environment code: ...**
**Task Description: ...**

**1. Implementation Details:**

(1) What packages are used in the code?

**2. Environment Structure**

(1) Observation Space:
1) What information does each agent observe?
2) Are the observations private to each agent, shared among all agents, or partially observable?
(2) Action Space:
1) What actions can each agent take?
2) Is the action space discrete, continuous, or a combination of both?
(3) State Space:
1) Does the environment maintain a global state?
2) If yes, how is it structured and is it accessible to all agents?
3) What variables make up the state space?

**3. Agent Interactions**

(1) Are the agents homogeneous or heterogeneous?
(2) How does the environment state evolve in response to agents' actions?
(3) Under what conditions can vehicles form platoons?

**4. Task-relevant information**

(1) What is the objective of the task?
(2) What variables may be associated with task objectives?
(3) What actions could an intelligent agent take that could be beneficial for completing a task?
(4) What indicators are related to tasks? Are these indicators positively or negatively correlated with the task?

First, let's take a deep breath, answering questions step by step.
Next, describe your reward function code in one sentence.
Finally, generate a reasonable reward function.

**LLM:**

**1. Implementation Details:**

(1) The code uses `numpy`, `networkx`, and `math` packages.

**2. Environment Structure**

(1) Observation Space:
1) Each agent observes its delay time, travel state, distance to the next hub, and information about nearby cars that might be potential platoon partners.
2) The observations are private to each agent and partially observable as they depend on the agent's current state and vicinity.
(2) Action Space:
1) Each agent can take actions corresponding to different speeds or waiting.
2) The action space is discrete, with a predefined set of possible speed actions.
(3) State Space:
1) Yes, the environment maintains a global state represented by the positions, travel states, and platoon partners of all vehicles.
2) It's structured as a dictionary holding each vehicle's status and is not directly accessible to other agents; each agent only has partial observations.
3) Variables include vehicle positions, travel states, platoon partners, normal travel time, delay time, and current edge in the path.

**3. Agent Interactions**

(1) Agents are homogeneous as they all operate with the same mechanics and objectives.
(2) The environment state evolves by updating vehicle positions, travel states, platoon statuses, and delay times based on the chosen actions.
(3) Vehicles form platoons when they choose the same action (speed) and are at the same position on an edge.

**4. Task-relevant information**

(1) The objective is to form as many platoons as possible by adjusting speeds.
(2) Variables include travel state, edge (current position), and platoon partner list.
(3) Actions that help align vehicles' speeds when they are close enough to form a platoon.
(4) The number of partners or vehicles in a platoon is positively correlated with the task, while delay time is negatively correlated. The reward function should incentivize forming platoons (increased platoon partners) and discourage unnecessary delays.

**5. Reward Function Code**

The reward function should incentivize forming platoons (increased platoon partners) and discourage unnecessary delays.
```python
...
```